



Analytical Problem Solving2

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QUESTIONS

Q1

Harshad number

10 marks



Q2

GCD

10 marks



Q3

sum to a target value

10 marks



Q4

Bitwise Operators

10 marks



Q5

binary search

10 marks



5/5 answered

Q3 sum to a target value

10 marks

Complete the missing lines in the Python program below. The program finds all pairs of elements in a list that sum to a target value. Fill in all seven blanks.

```
def find_pairs(lst, target):
```

```
    seen = _____ pairs = []
```

```
    for num in _____:
```

```
        complement = target - _____
```

```
        if complement in _____:
```

```
            pairs.append((complement, num))
```

```
            _____add(num)
```

```
    return _____
```

```
data = [int(x) for x in input("Enter numbers: ").split()]
```

```
t = int(input("Target: "))
```

```
result = _____(data, t)
```

```
print("Pairs:", result)
```

Your Answer

```
def find_pairs(lst, target):
```

```
    seen = set()
```

```
    pairs = []
```

```
    for num in lst:
```

Save Answer



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5/5 answered

Q4 Bitwise Operators

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

Your Answer

```
a = 0b1101
b = 0o25
c = 0x1A

result = (a & b) | c
```

Save Answer



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5/5 answered

Q1 Harshad number

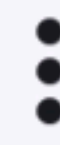
10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, $18 \% 9 == 0$. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
n=int(input())
s=0
t=n
while t>0:
    s+=t%10
    t//=10
```

 Save Answer



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Harshad number
10 marks

Q2

GCD
10 marks

Q3

sum to a target value
10 marks

Q4

Bitwise Operators
10 marks

Q5

binary search
10 marks

5/5 answered

Q2 GCD

10 marks

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

```
a = int(input("Enter first: ")) b = int(input("Enter second: "))
while b = 0: temp = a a = b b = temp % a
print("GCD is", a)
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 a=int(input())
2 b=int(input())
3 while b!=0:
4     temp=b
5     b=a%b
6     a=temp
7 print("GCD:",a)
```

Test Input

12
6

Output

GCD: 6

Pending manual review

Submitted